Cardiac Companion — Complete App Guide

Oct 3, 2026 · @Amin

Overview

Cardiac Companion (version 3.2) is a patient's own medicine record for heart patients in Ethiopia, built as one offline-first web app that runs in the phone's browser. The patient and their clinician enter the numbers; the app remembers them, counts down to what is due, reminds, and prints a summary for the next clinic visit.

Who it is for. People living with rheumatic heart disease, the most common cardiac condition among Ethiopian cardiac patients. That makes two regimens central: **warfarin** (blood thinner, monitored by INR blood tests) and **monthly benzathine penicillin injections** (to prevent rheumatic fever coming back). These are the two a paper card handles worst. Many users are older or unwell, so text is large and buttons are big.

What it is. A log, a reminder system and a visit-summary printer. It also offers an optional camera pulse measurement.

What it is not. It is not a measuring device or a decision-maker:

- There is no ECG, simulated or otherwise.
- It never works out or suggests a warfarin dose.
- It only compares an INR result with the range the doctor wrote down, and shows fixed warning-sign text. It interprets nothing else.
- The camera pulse is labelled an estimate, and its "uneven beats" note never names a condition.

Languages. English, Amharic (አማርኛ) and Afaan Oromo. Dates are shown in both the Ethiopian and the Gregorian calendar, and times in both the Ethiopian (day starts at 6 am) and the international clock.

Privacy. Everything stays on the phone, one record per person, with a one-file backup. Nothing is sent to any server.

Tech stack and project structure

The whole app is one HTML file plus two small scripts, written in plain HTML, CSS and JavaScript with no framework, no build step and no package manager. It is a Progressive Web App (PWA): it runs in the browser, works offline after the first visit, and can be added to the home screen like a native app.

File	Size	What it is
`index.html`	6,024 lines	The entire app: page layout, the stylesheet (design system), all screens, all three languages, all logic.
`ppg-engine.js`	1,601 lines	The camera pulse engine (photoplethysmography). Turns camera frames of a fingertip into a pulse rate, quality score and beat-to-beat variation. Loaded by `index.html`.
`sw.js`	72 lines	The service worker. Caches the app for offline use and handles taps on notifications.
`pulse-lab.html`	227 lines	A developer-only dashboard for testing the camera pulse engine live. Not part of the patient app.
`CHANGES.md`	152 lines	Release notes for versions 3, 3.1 and 3.2, including what must happen before real patients use it.
`tests/look.test.js`	72 lines	14 checks that the plain, flat design rules are kept.
`tests/ppg-synthetic.test.js`	366 lines	49 checks of the pulse engine on simulated signals.
`tests/pulse-monitor.mock.test.js`	281 lines	27 checks of the camera logic in a mock browser.

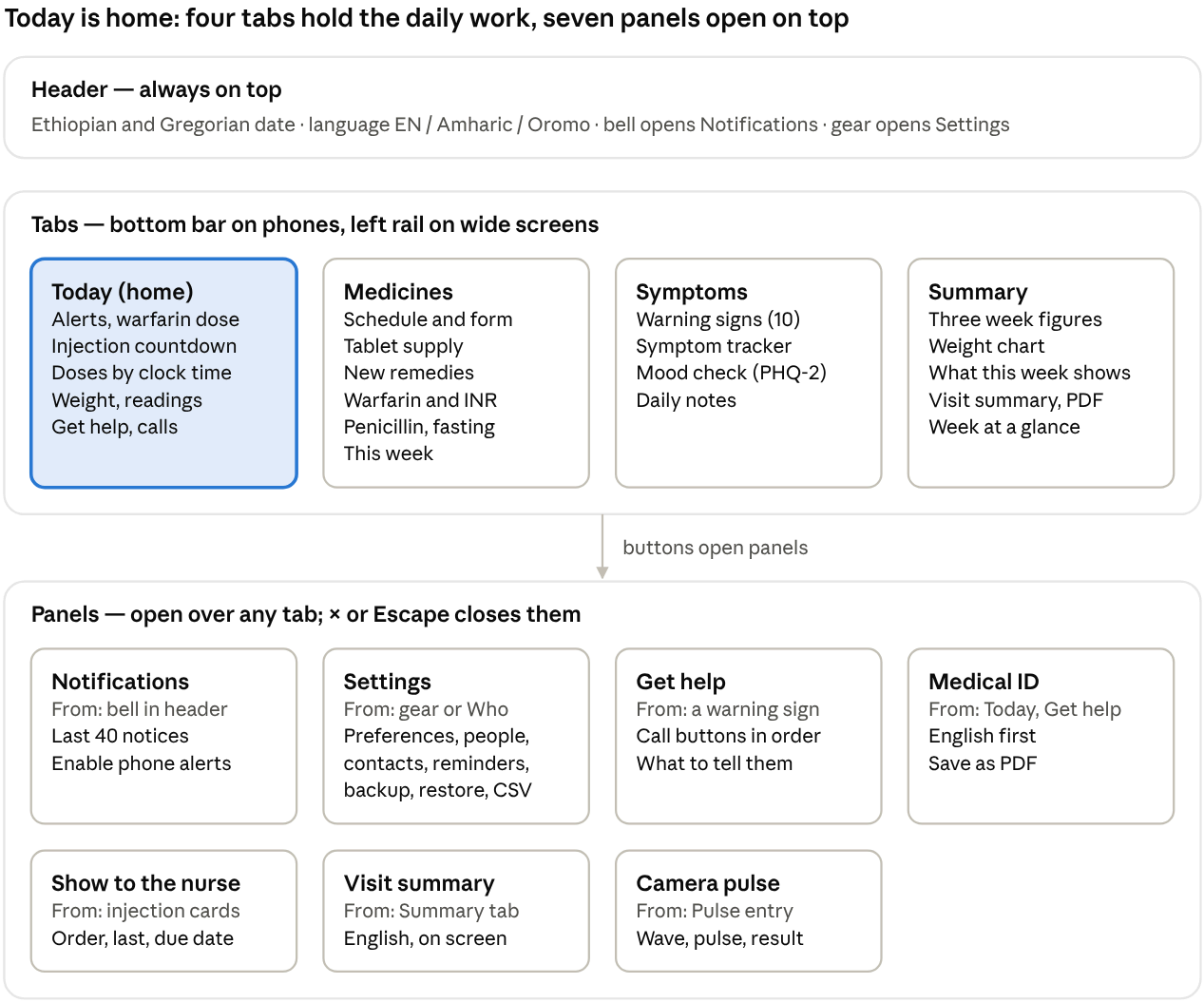
How it is put together

- **Deployment:** copy `index.html`, `sw.js` and `ppg-engine.js` into one folder on any static web host served over HTTPS (or `http://localhost`). All three are needed.

- **Web app manifest:** built at runtime inside `index.html` from three embedded PNG icons (192 px, 512 px and a 512 px maskable icon), so the app still ships as a single HTML file. Name "Cardiac Companion", short name "Cardiac", standalone display, portrait orientation, background and theme colour `#EEF1EE`.
- **Fonts:** Onest (for plain, easy-to-tell-apart numerals, which matters when the number is a dose) and Noto Sans Ethiopic (for correct Amharic script), both from Google Fonts. Without network on first load, the system font is used until the browser caches them.
- **Storage:** the browser's `localStorage` on the phone. No server, no account, no network calls apart from fonts.
- **Tests:** run with Node directly, for example `node tests/look.test.js`. No test framework is needed.

App flow and navigation

The app opens on Today every time; the bottom bar switches between four tabs, and buttons open full-screen panels on top that close back to where the person was.



navigation · header, 4 tabs, 7 panels

Alerts appear only on Today, and switching tabs scrolls the new tab to the top. A few buttons jump between places: *Warning signs* on Today goes to the Symptoms tab, the *Who* button and "Add your clinic's phone number" open Settings at the right section, and closing a panel returns focus to the button that opened it.

What happens when the app opens:

1. The page loads, from the phone's cache if there is no network.
2. The three languages' strings are merged, and the saved settings and list of people are read. On the very first open, one person is created, and an older version-2 record is upgraded if one is found.
3. Theme (saved, or the phone's), text size and language (saved, English by default) are applied.

4. Today's dose plan is made, any days missed since the last open are frozen with their plans, and a fasting period that has ended is closed.
5. Every screen is drawn and Today is shown with its alerts.
6. A notice is added: "Welcome — your medicine record is ready" on first use, or "Your record was updated" after an upgrade.
7. The 30-second clock starts, the service worker registers, and reminders are checked after 1.5 seconds.

Every screen explained

The app has a header, four tabs (Today, Medicines, Symptoms, Summary) and seven full-screen panels. A disclaimer sits at the foot of every tab: *"Not a medical device. This app records what you and your doctor enter — it does not measure, diagnose or decide anything. Always follow your clinician's guidance."*

Header (always visible)

Element	What it shows or does
Logo	A green square with a heartbeat line. Hidden on phones narrower than 420 px so the date fits.
Ethiopian date	Today in the Ethiopian calendar, large (for example "Meskerem 23"). Month names change with the language.
Gregorian date	Today in the Gregorian calendar with the year, smaller, underneath.
Language switch	EN, አማ (Amharic), OR (Afaan Oromo). Changes every word in the app at once and is remembered.
Bell	Opens the Notifications panel. A red dot appears when there are notifications.
Gear	Opens the Settings panel.

Today tab

Today is the home screen. It answers "what do I take now, and is anything wrong?"

Alerts strip (top of Today only). Shows only what needs attention now:

- Red: penicillin injection overdue ("N days late. Go to your health centre as soon as you can").
- Amber: INR test overdue or due today; "N doses due now — mark them below once taken"; "Medicine missed today"; "Running low" with each medicine's days left or "none left".
- One information card at a time, so the medicines never slide off the first screen: a detected fasting season ("It is the Lent fast (Abiy Tsom)" or "It is Ramadan" with *Yes, I'm fasting* / *No*), a clinic visit within two days, fasting advice while fasting mode is on, or a backup reminder (*Back up now* / *Later*, which snoozes it for 7 days).

Who button. Beside the "Today" title. Shows the person's name when a name is set or more than one person uses the phone. Tapping it opens Settings at "People on this phone".

Warfarin today (shown only when warfarin is switched on in Medicines). The dose is the biggest number on the page.

- Weekday label and the dose in mg (56 px), set by the doctor for that day of the week.
- Tablet pictures and words, such as "1½ × 5 mg tablet". If the dose cannot be made from the tablets: "This dose doesn't match your tablets — ask your pharmacist". If no strength is set: "Add your tablet strength to see how many tablets".
- "No dose today" when the doctor set 0 for that day; "No dose set" when it is blank.
- Buttons **Taken** and **Not taken**. Once marked: a state tag with the time ("Taken · 18:05") and an undo button.
- If not taken: one-tap reasons (Forgot, Ran out, Pharmacy had none, Couldn't afford it, Side effect, Fasting, Other), an optional free-text reason, and the line "Tell your doctor at your next visit — do not double the next dose."
- INR strip: the last four INR results, each with a drawn arrow (below range), tick (in range) or up arrow (above range), its date, and the target, for example "target 2.0–3.0".
- Next INR test: "Next INR test in 5 days · Tikimt 12 · 22 Oct", or a prompt to add the date on the Medicines page.
- A private pregnancy reminder when "Could this person become pregnant?" is answered Yes: "If your period is late or you might be pregnant, contact your heart clinic straight away. Do not stop warfarin on your own."

Penicillin injection (shown only when switched on in Medicines).

- A countdown box with the number of days. Neutral normally, amber when 3 days or fewer are left, red when due today or overdue.

- Wording: "Due in N days" with "Next due *date in both calendars*", "Injection due today", or "Injection overdue" with "N days late".
- **Record injection** opens a small form: the date of the injection (not in the future). If it was late, the app asks "That was N days after it was due. What happened?" with reasons: Forgot, Clinic had no penicillin, Health worker wouldn't give it, Couldn't travel or pay, Afraid of the pain, I was ill, Other.
- **Show to the nurse** opens the nurse card (see Panels).

Today's medicines. A "taken/planned" count (for example 2/5) sits in the header.

- Doses are grouped by clock time, each time shown in the international clock with the Ethiopian clock beside it ("08:00 · 2 in the morning (Eth.)").
- Each row: medicine name, strength, tablet pictures with "1 tablet" or "½ tablet", what it is for, then **Taken** / **Not taken**. Missed doses get the same reason chips and note field as warfarin.
- "When needed" medicines sit in their own group: "Taken 2× today, last at 14:00" or "Not taken today", a **Took one** button and undo.
- Empty states: "No medicines yet. Add one on the Medicines tab." or "No medicines due today."

Daily weight. "Weigh yourself each morning, after the toilet, before eating." A kilogram field (20–300 kg) and **Save**. Feedback underneath:

- First weight: "First weight saved — weigh yourself again tomorrow to start the trend."
- Steady: "Steady" with the last weight.
- Red alert when weight rose 2 kg or more against any of the previous three days: "Weight up 2.4 kg in 3 days. This can mean fluid is building up. Contact your clinician." This also goes to Notifications once per day.

Your readings. "Only if you measured it yourself with a cuff, thermometer or oximeter." Four tiles show today's latest value (with "×3" when there are several):

Tile	Entry	Accepted range (typo check only)
Pulse	beats per minute, typed or measured with the camera, plus "Did the beat feel regular?" Regular / Irregular / Not sure	20–250

Tile	Entry	Accepted range (typo check only)
Pressure	typed like 120/80	top 50–260, bottom 30–160, top higher than bottom
Oxygen	%	50–100
Temp	°C	30–45

Today's readings are listed with their time, each with a delete button. The ranges never judge a reading; they only catch slips like an oxygen of 850.

Get help. Call buttons appear for every number set in Settings: a red "Call ambulance (907)", "Call the clinic" and "Call *family member's name*". A link "Add your clinic's phone number" appears until one is set. Below are **Medical ID** and **Warning signs** (jumps to the warning signs on the Symptoms tab).

Medicines tab

Medicines is where the regimen is set up: ordinary medicines, warfarin, the penicillin injection, fasting and new remedies.

Schedule. The list of current medicines, each with name, strength, tablets per dose, how often, its times, what it is for, and the last change ("Changed 3 Oct: 40 mg → 80 mg"). Each row has an edit and a remove button. Removing asks first and keeps the medicine's past record; it simply stops being due from that day.

Add medicine / Edit medicine form:

Field	Choices
Name	Free text (required)
Strength	Free text, for example "40 mg"
Tablets each time	¼, ½, ¾, 1, 1½, 2, 2½, 3
How often	Every day, Every other day, Some days (then pick weekdays), Only when needed
Times	One or more clock times (not for "Only when needed"). "Add a time" proposes 12 hours after the last one. The Ethiopian clock is shown underneath.

Field	Choices
Tablets you have	Optional count, halves allowed; turns on supply tracking
What is it for?	Optional, shown under the name on Today

Every edit to dose, tablets, times or schedule is saved as a dated change and appears on the visit summary.

Supply left. "Counts down each time you mark a dose taken, halves included." For each medicine with a count (and warfarin, if it has one):

- "N tablets left" and "about N days", worked out from the schedule.
- A bar scaled to 30 days: green, amber when within the warning window (7, 14 or 21 days, set in Settings), red when none are left.
- A **Refill** field and plus button to add collected tablets.
- One notification when a medicine first enters the warning window and one when it runs out.

New medicine or remedy. "Antibiotics, pain medicines and herbal or traditional remedies can change how warfarin works. Note anything new here." (a gentler wording without warfarin). Type chips: Antibiotic, Pain medicine, Herbal or traditional, Other; an optional name and a date; **Save**. The last six entries are listed with delete buttons. With warfarin on, saving says "tell your warfarin clinic, and ask if you need an extra INR test".

Warfarin schedule (switch). "Enter exactly what your doctor wrote for each day. The app never works out a dose itself."

- A 7-day grid, Sunday to Saturday, one mg field per day, today's day highlighted, the tablet count under each day.
- A typo check: a dose over 20 mg in one day asks "Is that exactly what your doctor wrote?"
- Weekly total in mg and tablets a week.
- Tablet strength (1, 2, 3 or 5 mg), the time it is taken (default 18:00, with the Ethiopian clock), and warfarin tablets on hand.
- INR target range from the doctor, lowest and highest, between 1 and 5. Hint: "Often 2.0–3.0, or 2.5–3.5 with a mechanical valve — use exactly what your doctor wrote."
- Next INR test date.

- INR results: date and value (0.5–15; 8 or more asks for confirmation) and **Add**. Saving tells the person whether it is in range; an out-of-range result also adds a notification: "Make sure your clinic sees this result. Do not change your dose on your own."
- Time in target range over the last 6 months, as a percentage "from N results" (green at 65% or more).
- All INR results, newest first, each tagged below / in / above range and with the weekly dose in force at the time.
- Dose changes: the last five weekly totals with the date each started.

Penicillin injection (switch). "For rheumatic heart disease. Your clinic sets how often — every 2 to 4 weeks."

- Every 14, 21 or 28 days, or Other (7–60 days).
- Date of the last injection.
- "Your doctor's order, as written", for example "Benzathine penicillin 1.2 million units IM every 28 days".
- **Show to the nurse** button.
- "Last 12 months: 11 of 13 injections 85% · aim for at least 80%" (green at 80% or more).
- The last eight injections, each "On time" or "N days late" with its reason, and a delete button.

Fasting (switch). "Fasting changes when you eat and how much green food you eat — both can move your INR." When on: type chips (Orthodox fast, Ramadan, Other fast), a "Fasting until" date, and "While fasting" advice:

- "Take your medicines at the times your clinic agreed for fasting — ask them if you're not sure."
- With warfarin: "Don't skip warfarin, and keep leafy greens like gomen about the same each day." and "Ask your clinic whether you need an extra INR test."

The last four earlier fasts are listed with their dates. Fasting ends by itself the day after its "until" date.

This week. Seven day cells with "taken/planned": green when all were taken, amber when some, red when none, today outlined. Under it: "Not recorded this week: N doses. They count as not taken."

Symptoms tab

Symptoms records how the person feels and gives a fast route to help when something is dangerous.

Warning signs (the only card where red leads). Every sign is a button; tapping it records the sign with the time and opens the Get help panel.

Get help now if you have:	Call your clinic today if you have:
Black or bloody stool	Sore throat with fever
Vomiting blood	Unusual bruises or very heavy periods
Red or dark brown urine	Can't lie flat or wake up breathless
Bleeding that won't stop (nose, gums or a cut)	
Face drooping, arm weakness or trouble speaking	
Fainting or passing out	
Severe chest pain or can't breathe	

Symptom tracker. "Tap what you feel today, then say how bad it is." Eight standard symptoms plus any the person adds with **Add** (custom ones can be removed):

- Chest pain, Shortness of breath, Dizziness, Fatigue, Palpitations, Swollen ankles, Cough at night, Nausea.

Each ticked symptom opens its own detail: a "How bad?" slider from 1 to 10 and *At rest* / *When active*. Chest pain or shortness of breath at 7 or more shows "Severe chest pain or breathlessness is a warning sign." with a link to Get medical help now.

How you've been feeling (PHQ-2 mood check, every two weeks). "Over the last 2 weeks, how often have you been bothered by these?"

1. Little interest or pleasure in doing things
2. Feeling down, depressed or hopeless

Answers: Not at all (0), Several days (1), More than half the days (2), Nearly every day (3). A total of 3 or more shows "Please talk to your clinician — your answers suggest low mood. This is common with heart conditions and it can be treated. It is on your visit summary." Otherwise "Thank you — you'll be asked again in two weeks." The card then shows the date answered, the next check date and **Answer again**.

Daily notes. A free-text box ("How are you feeling today?"), **Save note**, and the last six notes with date, time and a delete button.

Summary tab

Summary shows the last week at a glance and builds the clinic visit summary.

Three figures across the top, all over the last 7 days:

Figure	Meaning
Adherence	Percentage of due doses taken. Unmarked past doses count as "not recorded" and the subtitle says how many.
Weight	Change in kg between the first and last weight of the week (needs two weights).
Symptom days	Days with at least one symptom, out of 7.

Weight chart. A line chart of daily weight in kilograms with a 7 days / 30 days switch. Fewer than two weights shows "No weights recorded yet this week."

What this week shows. Plain statements, each with a drawn tick (good), alert mark (needs attention) or info mark:

- "N of M doses taken this week", doses not recorded, missed doses or "No missed doses this week".
- Weight steady (under 1 kg change), up or down by N kg, or no weights.
- Days with symptoms, or none; warning signs reported this week.
- With warfarin: last INR with its date and range status; when the next INR test is.
- With penicillin: days until the injection, or days overdue.
- "Fasting mode is on."

Prepare for your visit.

- Last clinic visit and next clinic visit dates, and "Next visit in N days · *both calendars*".
- "Summary covers": Since last visit, 30 days or 90 days, with the exact date range. Without a last-visit date it falls back to 30 days and says "add your last visit date".
- "Add a copy in my language" (shown only in Amharic or Oromo) adds a second copy after the English one.

- **Doctor's summary (PDF, English)**, **Show on screen** and **Send to family** (see the visit summary feature below).

Week at a glance. A table with one row per day of the last 7: weekday, weight, doses taken/planned, number of symptoms.

Full-screen panels

Panels open over the app (on a wide screen, as a 420 px column on the right) and close with the × button or the Escape key.

Notifications. The in-app list of the last 40 notices, newest first, each with title, text, date and time; warnings have an amber border. At the top, until permission is given, "Turn on phone notifications for dose reminders" with **Enable**. At the foot, **Clear all**. Entries include: welcome on first use, record upgraded, missed doses, injection recorded with the next due date, out-of-range INR, weight gain, low or finished supply, and warning signs pressed.

Settings, in this order:

Section	Contents
Preferences	Appearance (Dark / Light); Text size (Normal / Large, about 14% bigger everywhere); Show times as (Both / 08:00 / Ethiopian); Warn me when a medicine will run out in (7 / 14 / 21 days)
People on this phone	Every person's record, with *Showing* on the current one, *Switch* on the others and a delete button (with a warning to back up first). The current person's name field. **Add a person** (a parent can keep a child's record; the new person copies the emergency contacts).
About this person	"Could this person become pregnant?" Yes / No / Prefer not to say, with why it matters for warfarin. On Yes: contraception used, and a "Planning a pregnancy" tick box.
Your conditions	Free text for conditions, allergies and past surgery (printed on the visit summary), and Heart valve: My own valves / Mechanical (metal) valve / Tissue valve / Not sure.

Section	Contents
Emergency contacts	Ambulance (907 by default), clinic or doctor, family member's name and phone.
Reminders	**Add reminders to my calendar**, a "Show medicine names in reminders" tick box, advice to re-add after changes; notification status (on / enable / not supported here) and **Send a test notification**.
Your data	"On this phone only — nothing is sent anywhere"; whether storage is protected from automatic clearing; last backup date (green within 30 days); **Back up now**, **Restore from a backup**, **Export data (CSV)**, **Add to home screen**.
About	"Cardiac Companion is a personal record, not a medical device."

Get help. Opened by a warning sign. Two levels:

- *Get medical help now* (red card): the sign pressed, then "This needs a doctor today. Don't wait to see if it settles — call an ambulance or go to the nearest hospital." Call buttons in the order ambulance, clinic, family.
- *Call your clinic today* (amber card): "Tell them what you noticed today. If it gets worse, go to a health centre." Call buttons in the order clinic, family, ambulance.
- With warfarin on, both add: Tell them: "I take warfarin, a blood thinner."
- If no clinic or family number is set: a prompt to add them in Settings. Buttons **Medical ID** and **Close**.

Medical ID. A card for emergency staff, English first with the person's language underneath. It lists the name, and only what applies:

- "Takes warfarin (blood thinner) — risk of bleeding." with the INR target and last INR.
- "Has a mechanical heart valve." or "Has a tissue heart valve."
- "Gets benzathine penicillin every N days (rheumatic heart disease)."
- The conditions text and the current medicines with strengths.

Below it are the call buttons and **Save as PDF**, which also prints the clinic, family and ambulance numbers.

Show to the nurse (titled Penicillin injection). A card to show at the injection room: "Benzathine penicillin G injection", "Every N days, as ordered by the patient's doctor", the doctor's written order, the last injection date, and the due date (with "N days late" if overdue). Without a written order it asks the person to add it.

Visit summary (Show on screen). The doctor's summary drawn as a white page, in English, with problem lines (out-of-range INR, late injections, warning signs, positive mood check, injections under 80%) in dark red.

Measure pulse with camera. Opened by **Measure with camera** in the Pulse entry. It shows the how-to text, a small camera preview, the pulse in beats per minute, a live pulse wave, a progress bar, a live hint, a "no light" note when the phone's light cannot switch on, and the result. Buttons: **Start**, then **Use this reading** and **Measure again**. A note at the foot says it is an estimate, not a medical device. (This panel was missing from the page until the fix described under Features.)

Features and logic

Every number the app shows comes from a small, fixed rule written in `index.html`; none of them decides treatment. These are the rules.

Doses and adherence

- **Daily plan.** Each day gets a frozen list of the doses that were due (one entry per medicine per time, plus warfarin when that day's dose is above 0). Days when the app was not opened still get their plan, because the schedule can only change while the app is open.
- **Which days a medicine is due.** Every day; every other day counted from its start date; chosen weekdays; or never for "only when needed". Not before it was added, not from the day it was removed.
- **Honest adherence.** Taken ÷ (taken + missed + not recorded). A past dose nobody marked counts as *not recorded*, never quietly dropped. Today's unmarked doses count as "due" once their time has passed and "upcoming" before that; neither counts against the person until the day is over.
- **Undo** returns a dose to pending and gives back any tablets it took from the supply.
- **Moving a time** carries a dose already marked today over to the new time.

- **Day rollover.** A clock runs every 30 seconds and whenever the app comes back on screen. At a new day it freezes yesterday, plans today, ends a fasting period that ran out and redraws.

Tablet supply

- Counted in tablets, quarters and halves included. Marking a dose taken subtracts that dose's tablets; for warfarin it subtracts the tablets that make that day's mg dose.
- Tablets per day: tablets each time × times a day; half that for every other day; × (chosen days ÷ 7) for some days; 0 for when needed. For warfarin: the week's mg ÷ tablet strength ÷ 7.
- Days left = tablets ÷ tablets per day, rounded down.
- Warnings fire once when days left reach the chosen window (7, 14 or 21 days) and once when the count reaches 0; refilling or changing the window resets them.

Warfarin and INR

- **Tablets for a dose:** dose ÷ strength, accepted only if it lands on a quarter tablet; otherwise "doesn't match your tablets".
- **Dose history:** every change to the 7-day schedule is stored with the date it started. Each INR result keeps the 7-day doses and weekly total in force on its test date.
- **INR status:** below the doctor's lowest number = below range, above the highest = above range, otherwise in range. No status without a target.
- **Time in therapeutic range (Rosendaal method):** draw a straight line between consecutive INR results and count the days the line sits inside the target. Gaps longer than 120 days are skipped, and it needs at least 7 days of line. Shown over the last 6 months; 65% or more is shown as good.
- **Typo checks:** daily dose over 20 mg asks for confirmation; INR must be 0.5–15 and 8 or more asks "Is that the number on your lab result?"; target values between 1 and 5 with the lowest below the highest; no future test dates.
- **The app never suggests a dose.**

Penicillin injection

- Next due = last injection + interval (14, 21, 28 or a custom 7–60 days).
- Lateness of each injection = days between its due date and the day it was given; late ones ask for a reason.

- **12-month completion:** injections given in the last 365 days ÷ injections expected since the first of them, with 3 days' grace before a due date counts. This is the ≥ 80% standard that rheumatic heart disease programmes use.

Health checks

Check	Rule	What the person sees
Weight gain	Latest weight is 2 kg or more above any weight from the 1–3 days before it	Red box and a notification: possible fluid build-up, contact your clinician
Weekly weight	Change between the first and last weight of the last 7 days	Steady under 1 kg, otherwise up or down by N kg
Severe symptom	Chest pain or shortness of breath rated 7 or more	Warning-sign note with a link to Get help
Mood (PHQ-2)	Two answers scored 0–3; total 3 or more of 6 is a positive screen; asked again after 14 days	"Please talk to your clinician"; flagged on the doctor's summary
Warning signs	Ten fixed signs in two levels	Logged with time; opens Get help

Calendars, clocks and fasting seasons

- **Ethiopian calendar:** Ethiopian New Year falls on 11 September (12 September when the next Gregorian year is a leap year). Months are 30 days, then Pagume. The Ethiopian year is the Gregorian year minus 7 from New Year, minus 8 before it.
- **Ethiopian clock:** hours count from 6 am, so 08:00 is "2 in the morning (Eth.)". Periods: morning 06–12, day 12–18, evening 18–24, night 00–06.
- **Dates people act on** (due dates, test dates, visits) are shown in both calendars, "Tikimt 12 · 22 Oct". Lists use the reader's likely calendar (Gregorian in English, Ethiopian in Amharic and Oromo). Oromo shows Gregorian dates as numbers, because its Ethiopian month names are the same words as its Gregorian ones.
- **Days are keyed by the phone's local date**, never UTC, because in Ethiopia (UTC+3) UTC is still "yesterday" until 03:00.
- **Abiy Tsom (Orthodox Lent):** the 55 days before Orthodox Easter, which is computed with the Julian rule and shifted 13 days to the Gregorian calendar.
- **Ramadan:** the 9th month of the browser's Islamic (Umm al-Qura) calendar.

- When a season is on and the person has medicines, Today asks once per season whether they are fasting. "Yes" turns on fasting mode until the season's last day.

People on one phone

Each person has a separate record (medicines, readings, summary). Switching saves the current record and loads the other. Reminders run for every person on the phone, with the person's name in the title when there is more than one.

Reminders

A web page cannot ring an alarm while it is closed, so reminders come in two layers.

Layer 1 — phone notifications while the app is open or was used recently. The reminder engine runs every 30 seconds and when the app returns to the screen:

- Each unmarked dose is announced once at its time (only if no more than 3 hours late) and once more an hour later if still unmarked (only if no more than 4 hours late). Never hours after the fact.
- Doses due at the same time share one notification: "Medicines at 08:00 — 3 to take, open the app to mark them". Medicine names appear only if the person ticked "Show medicine names in reminders".
- Once a day after 07:00, a "Coming up" notification for: the penicillin injection (due tomorrow, today or late), the INR test (tomorrow, today, or up to 7 days overdue) and a clinic visit (today or tomorrow).
- It runs for every person on the phone. Tapping a notification brings the app forward or reopens it.
- The app never notifies about something the person just did (no "Dose recorded" pop-ups).

Layer 2 — the phone's own calendar, which rings even when the app is closed and offline. "Add reminders to my calendar" builds a calendar file (.ics) and hands it to the share sheet or downloads it:

- One repeating event per dose time: daily, every 2 days, or on the chosen weekdays; warfarin on the days its dose is above 0. Alarm at the time.
- One-off events for the next penicillin injection (09:00), the next INR test (08:00) and the next clinic visit (08:00), each with an alarm the day before and at the time.

- Titles never include doses, because a dose in a calendar goes stale the day the doctor changes it. Medicine names appear only if chosen, because calendars can sync to the cloud. After any change the person re-adds the file and deletes the old events.

Visit summary

One structured summary is rendered three ways: a PDF, an on-screen page, and plain text for sharing. Its period is "since last visit", 30 days or 90 days.

Doctor's summary is always in English (clinicians write their notes in English), optionally followed on a new page by a copy in Amharic or Oromo. It contains, when present:

1. Title, the person's name, and the period in both calendars.
2. Conditions: the free text, heart valve, and pregnancy possibility with contraception.
3. Medicines with strength, tablets, times and schedule; changes in the period; medicines stopped; dose totals (taken, missed, not recorded, % taken); reasons for missed doses; "when needed" counts.
4. Warfarin: the 7-day doses and weekly total, tablet strength, time, target; warfarin dose totals and the dates missed with reasons; INR results from 60 days before the period onward with status and weekly dose; time in range over 6 months; dose changes; next INR test.
5. Things that can change INR: new medicines and remedies, and fasting periods.
6. Penicillin: interval, last injection, due date, lateness; 12-month completion; late injections with reasons; the doctor's order.
7. Warning signs pressed, with date and time.
8. Symptoms per day with severity and rest/active.
9. Mood (PHQ-2) scores.
10. Weights with the change; readings for up to 14 days.
11. Daily notes (last 8).
12. The not-a-medical-device disclaimer.

Family update ("Send to family") is shorter: it leaves out conditions and pregnancy, mood and notes. It goes to the phone's share sheet (for example Telegram), or is copied to the clipboard to paste into a message.

How the PDF is made. PDF's built-in fonts cannot encode Ethiopic script, and embedding a font would add megabytes. So each A4 page is drawn on a canvas with the fonts the browser already has, saved as a JPEG, and placed in a PDF written by the app itself. File name: `cardiac-summary-<name>-<date>.pdf`.

Exports, backup and restore

Action	File	What it holds
Back up now	`cardiac-companion-backup-<date>.json`	Every person on the phone, their whole record. Shared (for example to Telegram Saved Messages) or downloaded.
Restore from a backup	reads that JSON	Checks it is a Cardiac Companion backup, shows its date and names, asks for confirmation, then replaces everything on the phone and reloads.
Export data (CSV)	`cardiac-companion-<name>-<date>.csv`	One row per day: date, doses planned, taken, missed, not recorded, warfarin mg, warfarin status, INR, weight, symptoms, readings, warning signs, fasting.
Medical ID → Save as PDF	`medical-id-<name>-<date>.pdf`	The medical ID lines and phone numbers.
Add reminders to my calendar	`cardiac-reminders-<name>-<date>.ics`	The calendar events above.

Backup reminder. Appears on Today 3 days after the record starts if there has never been a backup, then whenever the last backup is more than 30 days old. *Later* snoozes it for 7 days.

Protected storage. After the first thing the person saves, the app asks the browser to keep its data from automatic clearing. Settings shows whether this was granted.

Screen behaviour worth knowing

- A section is never redrawn while the person is typing in it; the update waits until they leave the field.
- If the app is open in two tabs, a change in one shows in the other.
- Large text zooms the whole app by about 14%, keeping every layout in proportion.
- Dark mode follows the phone's setting on first use and can be switched in Settings.

Camera pulse (ppg-engine.js)

The camera pulse estimates the pulse rate from a fingertip held over the phone's back camera and light. It is an optional estimate, saved like a typed pulse and marked " (camera)".

How it works in the app. In the Pulse entry, a **Measure with camera** button opens a panel. The person rests, covers the camera and light with a fingertip and holds still for about a minute. Live hints guide them: "Cover the camera and light with your fingertip.", "Hold still…", "Press a little lighter.", "Weak signal — warm your hand and cover the light fully.", "Good — keep still." A wave and a progress bar show while it measures. At the end:

- The pulse in beats per minute, and **Use this reading**, which puts the number in the pulse field where it can still be edited. Typing a different number drops the camera details.
- Beat-to-beat variation (RMSSD and SDNN, in ms) only when the engine judges it reliable; otherwise a plain reason: beats were uneven, not enough clean beats, too many beats disturbed, or the signal was not precise enough.
- If beats were uneven: "Your beats were uneven during this reading. This can have many causes. If you see this again, or feel dizzy or breathless, tell your nurse or doctor." It pre-selects "Irregular" in the rhythm question, which the person can change. It never names a condition.
- Always: "This is an estimate from your phone's camera, not a medical device." Phones whose light cannot be switched on get "Measure next to a bright lamp or window."
- The saved reading keeps its details: method, RMSSD, SDNN, whether variation was reliable, beats used, signal quality, uneven or not, and the engine version.

How the engine works (version 1.0.0), per camera frame at about 30 a second:

1. Average the red, green and blue of the centre of the frame, and how much red is clipped.
2. Decide whether a fingertip is on the lens (red-dominant and bright enough).
3. Re-sample the red values onto an exact 30 Hz grid using each frame's capture time.
4. Band-pass filter 0.7–3.5 Hz (42–210 bpm) forwards and backwards over an 8-second window, so peaks are not delayed.
5. Find peaks with Elgendi's two-moving-average detector, refined to between frames.
6. Compare each pulse with a template of its neighbours; set aside wrong shapes, weak blips and movement.

7. Accept each beat-to-beat interval only if it is within ±25% of recent intervals or ±15% of the one before; merge split beats and detect missed ones.
8. Score quality from contact, pulse shape, regularity, timing precision, movement, glare and the share of accepted intervals.

It also turns the phone's light on (trying other back cameras if needed), lowers exposure when red is clipped, keeps the screen awake while measuring, and gives up after 3 minutes without enough clean signal. A reading needs at least 10 clean intervals (or 15 beats when the rhythm is uneven).

Fixed on 3 Oct 2026 (fixed `` **, not yet merged).** Until this fix, the button and the whole pulse panel were missing from `index.html` (also in v3.1), although the code referred to them. Tapping any reading tile threw "Cannot set properties of null (setting 'hidden')", the pulse rhythm buttons never appeared and the camera could not be reached. The fix adds the button and panel in the app's plain style. It was checked in Chromium with a simulated camera: no errors, rhythm buttons shown, panel opens, starts and closes, and "Use this reading" saves "Pulse 72 (Irregular) (camera)". The engine can also be tried on its own in `pulse-lab.html`.

Before real patients use it (from CHANGES.md): clinical sign-off on the wording and on whether variability should be shown at all; real-world validation against a pulse oximeter or a 60-second manual count on the phones patients use (for example TECNO, itel, Infinix, Samsung A-series), including dark skin, cold hands and atrial fibrillation; and a native speaker's review of the Amharic and Oromo camera texts. In atrial fibrillation some beats never reach the fingertip, so the camera, like a finger on the wrist, can count fewer beats than the heart makes.

Health and educational content

All health wording in the app is fixed text that follows standard warfarin and rheumatic heart disease patient leaflets; nothing is generated. The warning signs are listed under the Symptoms tab above. The rest, by topic:

Topic	Exact wording in the app (English)	Where it shows
Warfarin dose	"This is the dose your doctor set for this day of the week."	Warfarin card

Topic	Exact wording in the app (English)	Where it shows
Warfarin setup	"Enter exactly what your doctor wrote for each day. The app never works out a dose itself."	Warfarin schedule
Missed warfarin	"Tell your doctor at your next visit — do not double the next dose."	After "Not taken", and in Notifications
INR target	"Often 2.0–3.0, or 2.5–3.5 with a mechanical valve — use exactly what your doctor wrote."	Warfarin schedule
INR out of range	"Make sure your clinic sees this result. Do not change your dose on your own."	After saving an INR, Notifications
INR test due	"Your INR test is due. Take your warfarin record with you."	Calendar reminder
Things that change INR	"Antibiotics, pain medicines and herbal or traditional remedies can change how warfarin works. Note anything new here."	New medicine or remedy card
Fasting	"Fasting changes when you eat and how much green food you eat — both can move your INR." plus the three "While fasting" tips (agreed times; don't skip warfarin, keep leafy greens like gomen steady; ask about an extra INR test)	Fasting card, Today alert
Pregnancy	"Warfarin can harm an unborn baby…" and "If your period is late or you might be pregnant, contact your heart clinic straight away. Do not stop warfarin on your own."	Settings, warfarin card (private)
Penicillin	"For rheumatic heart disease. Your clinic sets how often — every 2 to 4 weeks." / "Go to your health centre for your penicillin injection."	Injection setup, calendar

Topic	Exact wording in the app (English)	Where it shows
Injection late	"N days late. Go to your health centre as soon as you can."	Today
Weight	"Weigh yourself each morning, after the toilet, before eating." / "This can mean fluid is building up. Contact your clinician."	Daily weight
Readings	"Only if you measured it yourself with a cuff, thermometer or oximeter."	Your readings
Get help now	"This needs a doctor today. Don't wait to see if it settles — call an ambulance or go to the nearest hospital."	Get help panel
Call clinic today	"Tell them what you noticed today. If it gets worse, go to a health centre."	Get help panel
What to say	"I take warfarin, a blood thinner."	Get help panel
Severe symptom	"Severe chest pain or breathlessness is a warning sign."	Symptom tracker
Low mood	"Your answers suggest low mood. This is common with heart conditions and it can be treated."	Mood check
Low supply	"About N days left. Plan your pharmacy trip." / "You have run out. Collect more as soon as you can."	Notifications
Backup	"If this phone is lost or its browser data is cleared, everything here is gone. A backup takes a few seconds."	Today alert
Disclaimer	"Not a medical device. This app records what you and your doctor enter — it does not measure, diagnose or decide anything."	Every tab, every summary

Languages. Every string exists in English, Amharic and Afaan Oromo: 92 interface labels and lists, and 494 messages, with no gaps between languages. The Amharic and Oromo strings added in version 3 (warning signs, pregnancy note, mood questions especially) still

need a bilingual health worker's review, and the camera texts are marked as drafts. A few version-2 strings (stress level, symptom context, date format) remain in the file but are no longer shown.

Sign-off still needed (from CHANGES.md): clinical approval of the fixed wording (warning signs, fasting advice, camera wording) before real patients use the app.

Data, storage and services

All data lives in the browser's `localStorage` on the phone; the app has no server, no account and sends nothing anywhere. The only network requests are for the two Google fonts and the app's own files.

Storage keys

Key	Holds
`cc-state-v3:<person id>`	One person's whole record (below)
`cc-profiles`	The list of people on the phone and which one is showing
`cc-global`	Last backup date, backup snooze date, whether persistent storage was asked, large text on/off
`cc-lang`	Chosen language (en, am, or)
`cc-theme`	light or dark
`cc-state-v2`	The version-2 record, left untouched as a fallback after upgrading
`ppg-camera-id`	Which back camera had a working light (camera engine)

One person's record (version 3)

Field	What it stores
`meds`	Each medicine: id, name, strength, purpose, times, schedule (type, weekdays, start), tablets per dose, supply, date added, date removed, change history, low/out warning flags
`dayPlan`, `planDay`	For each date, the frozen list of doses that were due; the last day planned
`doseLog`	For each date and dose: taken or missed, time marked, reason, note, tablets deducted
`prnLog`	"When needed" doses taken, with time and tablets deducted
`legacyAdherence`	Daily taken/total tallies carried over from version 2
`warfarin`	On/off, 7 daily doses, tablet strength, time, supply, warnings, INR target, next INR date, dose history
`inrLog`	Each INR: date, value, the 7 doses and weekly total in force
`injection`	On/off, interval, last date, doctor's order, history (date, due date, days late, reason)
`weightLog`	Date → kg
`vitals`	Date → readings (time, kind, value, rhythm, camera details if any)
`symptomLog`, `legacySymptomCount`, `customSymptoms`	Each day's symptoms with severity and rest/active; old day counts; the person's own symptoms
`redFlags`	Warning signs pressed (date, time, which sign)
`notes`	Daily notes (date, time, text)

Field	What it stores
`newMeds`	New medicines and remedies (date, type, name)
`fasting`	On/off, type, until, past periods, seasons dismissed
`mood`	PHQ-2 answers by date
`about`	Pregnancy answer, contraception, planning, heart valve
`clinicalInfo`	Conditions, allergies, surgery text
`contacts`	Ambulance (907), clinic, family name and number
`visits`	Last and next clinic visit
`settings`	Low-supply days (14 by default), clock style (both), names in reminders (off)
`notifications`	The last 40 in-app notices
`fired`	Which reminders were already sent, kept for 7 days

Every stored record is checked and repaired on load, so a missing or damaged field falls back to its default instead of breaking the app.

Upgrade from version 2

On first open of version 3, an existing version-2 record is converted: medicines (with their time and today's status), daily tallies, warfarin doses and log, INR results, injection settings and history, weights, readings, symptoms, notes, conditions and the emergency contact (moved to the clinic number). A notice says "Your record was updated". Exact per-dose history starts from the upgrade day.

Browser features the app uses

Feature	Used for	Asked for when
Notifications	Dose reminders and "Coming up" notices	The person taps Enable
Service worker (`sw.js`)	Working offline; showing notifications; reopening the app from a notification	Automatically
Persistent storage	Protecting the record from automatic clearing	After the first thing saved
Camera and light	Camera pulse	Only when a measurement starts
Screen wake lock	Keeping the screen on while measuring	During a measurement
Share sheet / clipboard	Sending the family update, backup, calendar file	When the person taps the button
Phone dialler (`tel:` links)	Call buttons	When tapped
Install prompt	Add to home screen	When the person taps it (otherwise a tip to use the browser menu)

Service worker (`sw.js`)

- Cache name `cardiac-companion-v6`; older caches are deleted when a new version activates.
- On activation it caches the open page and `ppg-engine.js`, so even the first visit works offline afterwards.
- Network first, cache as fallback, for the app's own files and Google Fonts. Only GET requests are handled.
- Tapping a notification focuses an open app window or opens a new one.
- It cannot schedule notifications itself; reminders come from the open page and the calendar file.

Privacy and law

Nothing leaves the phone except what the person chooses to share or download. CHANGES.md notes that if a server is ever added (SMS, Telegram, sync), Ethiopia's Personal Data Protection Proclamation 1321/2024 requires explicit consent for health data and local storage of personal data.

Design system and assets

The look is deliberately plain, so the app reads like a medical record rather than an app template (version 3.2). The rules sit in a comment at the top of the stylesheet and are enforced by `tests/look.test.js`.

Rules

- White cards with a 1 px border on a pale green-grey page. No shadows, gradients, pill shapes, glows or decorative dots.
- One green for actions, red only for warning signs and danger, amber for "soon" and "low".
- Corners: buttons and fields 8 px, cards 12 px, state tags 6 px.
- Nothing moves under the finger: colours change in 120 ms, panels appear in place, and motion is switched off entirely when the phone asks for reduced motion.
- Counts ("2/5", the weekday) are plain text; only states (taken, missed, below / in / above range) get an outlined tag.
- Icons are drawn, never text characters such as ✓ or ×.
- Controls are at least 48 px tall (main buttons and fields) and 44 px (small buttons, chips, icon buttons) on phones, because many users are older or unwell. With a mouse on a screen 720 px or wider they shrink to 40 px and 36 px.

Colours

Token	Light	Dark	Used for
`--canvas`	#EEF1EE	#0D1311	Page background
`--surface`	#FFFFFF	#161E1B	Cards, fields
`--surface-2` / `-3`	#F4F6F5 / #E7ECE9	#1C2622 / #26312D	Insets, pressed states

Token	Light	Dark	Used for
`--line` / `--line-strong`	#DCE2DF / #BFC9C4	white at 9% / 20%	Borders
`--ink-1` / `-2` / `-3`	#14201C / #435049 / #5B6862	#E8EFEC / #B2BFB9 / #8F9C96	Text, secondary text, hints
`--accent`	#1D5C54	#7DCABA	Actions, chosen tab, chart line
`--danger`	#B3261E	#F08F85	Warning signs, overdue, ambulance button
`--warn`	#D39A1C	#EDC05A	Soon, low supply, alerts

Each colour also has soft, line and ink variants for backgrounds, borders and text on those backgrounds.

Type

- Onest for Latin text and numbers (plain numerals: no slashed zero, distinct 1/7, 5/6, 3/8), Noto Sans Ethiopic for Amharic, then the system font.
- Body 15 px; page titles 24 px; card titles 17 px; the warfarin dose and the camera pulse 56 px; weights up to 700 only. Numbers use tabular figures so columns line up.

Layout

- **Phone:** one column up to 460 px wide, header on top, tab bar at the bottom (the chosen tab has a bar along its top edge). Respects notches and home bars.
- **Tablet and desktop (720 px and wider):** the tab bar becomes a 104 px rail on the left (chosen tab tinted), cards sit in two columns, and panels open as a 420 px column on the right.
- **Large text:** the whole app zooms by 1.14.
- **Dark mode:** a full second set of colours; follows the phone on first use.

Icons and pictures

Asset	Where
Heartbeat line	Logo and Today tab; also the Pulse tile

Asset	Where
Capsule, heart, bar chart	Medicines, Symptoms and Summary tabs
Bell, gear, close (× drawn)	Header and panel headers
Button icons (drawn as masks so translated labels never wipe them)	Plus, check, PDF, share, phone, download, upload, phone-device, calendar, ID card, eye, warning triangle, bell, syringe (nurse), camera
State marks	Alert circle, info circle, tick, down arrow, up arrow, "open" chevron
Reading tiles	Heartbeat (pulse), wave (pressure), drop (oxygen), thermometer (temperature)
Tablet pictures	A tablet drawn as two halves with a score line; ½ and ¼ show the missing part dashed. More than four whole tablets show as one picture "×N". Always beside words, never instead of them.
App icons	192 px, 512 px and a 512 px maskable PNG, embedded in the HTML as base64; also used as the browser tab icon, the home-screen icon and the notification icon

Accessibility

- Tabs are a tab list, panels are modal dialogs, toggles are switches, symptom tiles are checkboxes; all labels, including screen-reader labels, are translated.
- Focus moves to a panel's close button when it opens and back to the button that opened it when it closes; Escape closes the top panel; Enter saves weight, readings and INR.
- INR arrows carry the status in words for screen readers; toasts and camera hints are announced.
- The page language switches with the app (en, am, om for Oromo) so screen readers pronounce it correctly.

Configuration, build, tests and tooling

There is no build step and no configuration file: the app is edited directly and deployed by copying files. All 90 automated checks pass as of 3 Oct 2026.

Deploy

1. Copy `index.html`, `sw.js` and `ppg-engine.js` into one folder on a static web host served over HTTPS (or `http://localhost` for testing). All three are needed.
2. Leave out `pulse-lab.html`, `tests/` and `CHANGES.md`; they are for developers.
3. When releasing a new version, bump the cache name in `sw.js` (now `cardiac-companion-v6`) so phones drop the old copy. Because the worker fetches from the network first, changes show up on the next online visit anyway.

Version markers: app version 3.2 (comments in `index.html` and `sw.js`), service-worker cache v6, pulse engine 1.0.0, record format v3.

Tests

Run each with Node; no install needed. Each exits with code 1 if any check fails.

Command	Checks	What it proves
`node tests/look.test.js`	14	The flat look stays: no gradients, pill or round buttons, shadows, motion under the finger, keyframe or slide-in animation, all-caps, extra-bold type, or text characters as icons; controls 48/44 px; corners 8 and 12 px.

Command	Checks	What it proves
`node tests/ppg-synthetic.test.js`	49	The pulse engine on simulated fingertip signals with known answers: normal, slow (40–45 bpm) and fast (140 bpm) rhythms, weak and noisy signals, a missed beat and a false peak, uneven (AF-like) rhythm, movement and a lifted finger, premature beats, strong breathing variation, 15 fps cameras and coarse clocks, a timed 30-second reading, a camera pointed at a room, reliability across 56 recordings, the uneven flag across 32 recordings, and the zero-phase filter.
`node tests/pulse-monitor.mock.test.js`	27	The camera layer in a mock browser with three cameras: finding and remembering the camera with a light, torch on, screen kept awake, the state sequence, release of camera and wake lock, a refused permission, switching away mid-reading, glare and exposure stepping, a browser that cannot switch the light on, and older animation-frame timing.

What the tests do not cover: the app screens themselves (beyond the look rules), and real cameras, skin and patients.

Pulse lab (`pulse-lab.html`)

A developer page that shows the pulse engine live on a phone. Open it next to `ppg-engine.js` over HTTPS.

- Header: live pulse with a flashing beat dot, status line and the camera preview.
- Measurement: duration (30 s, 60 s, 2 min, Continuous), **Start**, **Stop**, progress bar.
- Signal quality: level, score, hint, and a no-light note.
- Pulse wave (green: beat used; amber: set aside) and a chart of beat-to-beat intervals with last, instant and average values.

- Heart-rate variability: RMSSD, SDNN, pNN50, a reliability tag with the reason, and the uneven-beats note.
- Counts: beats, set aside, intervals rejected, beat timing precision.
- Camera: which camera, light, frame rate, clock source, exposure.
- Result text, **Download JSON**, **Download intervals (CSV)**, and a live log of every engine event.
- Footer: "Not a medical device. These numbers are estimates from a phone camera and have only been tested on simulated signals."

Before real patients use it

From CHANGES.md, still open:

- [ ] Bilingual health-worker review of all Amharic and Afaan Oromo text, especially warning signs, the pregnancy note, the PHQ-2 questions and the draft camera texts.
- [ ] Clinical sign-off on fixed wording (warning signs, fasting advice) and on the camera feature, which changes the app's "not a measuring device" charter.
- [ ] Real-world camera validation against a pulse oximeter or a 60-second count on TECNO, itel, Infinix and Samsung A-series phones, including dark skin, cold hands and atrial fibrillation.
- [ ] True alarms while the app is closed need a thin native wrapper (for example Capacitor Local Notifications), tested on TECNO, itel and Infinix phones.
- [ ] If a server is ever added: explicit consent and local storage of health data under Proclamation 1321/2024.

Version history

Date	Version	Change
3 Oct 2026	3.2 fix	Camera pulse button and panel added back (fixed , not yet merged)
29 Sep 2026	3.2	Plain, flat look; plural wording fixed; weight chart dates fixed; look test added

Date	Version	Change
29 Sep 2026	3 and 3.1	Replaced with the uploaded v3.1 package. Version 3: reminders, backup, INR range and time in range, warning signs, contacts, flexible medicine schedules, penicillin tracking, fasting, new remedies, pregnancy prompt, visit summary, profiles, both calendars and clocks. Version 3.1: camera pulse, `ppg-engine.js`, `pulse-lab.html` and the engine tests
29 Sep 2026	2	Rebuilt as a heart-health companion (record format `cc-state-v2`)
11 Sep 2026	1	First upload: `cardiac_companion_cleaner.html` (renamed to `index.html`) and `sw.js`, record format `cc-state-v1`

File-by-file index

A map of where everything lives, so any part of the app can be found by line. Line numbers are for `index.html` with the camera fix (6,024 lines).

`index.html`

Lines	Part	Contents
1–51	Head	Page settings, theme colour, manifest link, a comment on what the app is and is not, Google Fonts
52–1124	Stylesheet	Design rules comment, colour tokens (light and dark), base, drawn icons, header, cards, tags, buttons, fields, switch, Today, Medicines, Symptoms, Summary, alerts, nav, panels, toast, hover, desktop rail, additions (large text, tablets, INR flags, warning signs, mood, help, ID card, preview), camera pulse

Lines	Part	Contents
1129–1157	Header	Logo, both dates, language switch, bell, gear
1163–1316	Today tab	Alerts strip, warfarin card, injection card, today's medicines, weight, readings (with the camera button), get help
1319–1405	Medicines tab	Schedule, supply, new medicine or remedy, warfarin, penicillin, fasting, this week
1408–1472	Symptoms tab	Warning signs, tracker, mood check, notes
1475–1552	Summary tab	Three figures, weight chart, insights, visit preparation, week table; disclaimer
1556–1573	Nav	Today, Medicines, Symptoms, Summary
1578–1776	Panels	Notifications, Settings, Get help, Medical ID, Show to the nurse, Visit summary, Camera pulse; hidden restore file input; toast
1778	Engine	Loads `ppg-engine.js`
1779–1832	Manifest script	Three embedded PNG icons, web app manifest built at runtime, favicon, early theme
1834–1884	Ethiopian calendar	Month names in three languages, Gregorian → Ethiopian conversion
1885–2296	Strings (v2)	`T.en`, `T.am`, `T.or`: labels, weekday names, the 8 symptoms, messages
2300–2950	Strings (v3)	`T2.en`, `T2.am`, `T2.or`: everything added in version 3, merged at start-up
2954–3031	State and helpers	Storage keys, globals, date keys, number parsing, "is the person typing" check

Lines	Part	Contents
3032–3158	Language, dates, clocks, fasting seasons	`L()` lookup with plural fix, both calendars, Ethiopian clock, Orthodox Easter, Ramadan
3159–3345	Record and storage	Default record, repair on load, v2 → v3 upgrade, profiles, persistent storage
3346–3600	Core rules	Dose plans and adherence tally, supply, warfarin and INR (time in range), penicillin completion, symptoms and mood, fasting
3601–3670	Shared pieces	Drawn icons, reason lists, tablet pictures, "don't redraw while typing"
3671–4137	Today logic	Warfarin, injection, medicines, weight, readings, camera pulse, get help
4138–4556	Medicines logic	Schedule and form, supply and refill, warfarin and INR, injection setup, fasting, new remedies, week view
4557–4702	Symptoms logic	Warning signs and help panel, tracker, mood, notes
4703–4850	Summary logic	Figures, weight chart, insights, visit card, week table
4851–5209	Alerts, settings, people, ID, notifications, panels	Today alerts, all Settings sections, profiles, medical ID and nurse card, notification list, panel and tab switching, language switch, full redraw
5210–5336	Event wiring	One table for every button (`data-act`), every keystroke (`data-in`) and every change (`data-ch`); Escape and Enter keys
5337–5534	Visit summary	Summary blocks for doctor and family, plain text, on-screen preview, share
5535–5698	PDF	Canvas pages and a small PDF writer; doctor's summary and medical ID PDFs

Lines	Part	Contents
5699–5808	CSV and calendar	Daily CSV; .ics calendar file
5809–5886	Backup and notifications	Backup and restore; phone notifications and permission
5887–5983	Reminders, clock, install	Reminder engine, 30-second clock and day rollover, install prompt
5984–6024	Start	Merge strings, load data, theme, text size, language, welcome notice, timers, service worker

`ppg-engine.js` (1,601 lines)

Lines	Part
1–92	Documentation: what it does, the 8-step pipeline, what the numbers mean, uneven rhythms, data hooks, not-a-device note
93–173	Setup and every tunable default (filter band, window, beat shape limits, interval tolerances, HRV reliability limits, contact and glare thresholds, session length, camera size, torch, exposure, wake lock)
174–401	Small maths, filters (band-pass, zero-phase), peak detection, periodicity, beat templates
402–767	`RRAnalyzer`: interval acceptance and rejection, merges, missed beats, HRV metrics, uneven-rhythm test
768–1172	`SignalProcessor`: frames in; beats, intervals, quality and the final summary out (runs in Node)
1173–1504	Camera: open the back camera, find the light, sample frames, adapt exposure, wake lock; `PulseMonitor` with its events

Lines	Part
1505–1601	Drawing helpers (`drawWave` , `drawTachogram`) and the public `PPG` object

Other files

File	Contents
`sw.js`	Cache name; install (activate at once); activate (drop old caches, cache open pages and the engine, take control); fetch (network first, cache fallback, own files and fonts only); notification click (focus or open the app)
`pulse-lab.html`	Developer dashboard for the engine (see Pulse lab above)
`CHANGES.md`	Release notes for 3.2, 3.1 and 3, each with what changed, what was fixed or added, and what must happen before patients use it
`tests/look.test.js`	14 design-rule checks on the stylesheet and markup
`tests/ppg-synthetic.test.js`	15 simulated scenarios, 49 checks of the engine
`tests/pulse-monitor.mock.test.js`	7 camera scenarios in a mock browser, 27 checks

The repository has no `README` , no `package.json` , no licence file and no CI workflow.